Discharging H-diag II: the off-diagonal Bogoliubov Hessian is the

TECT verification-first repository · claim B2-PROPA-HLAYER

first issued 2026-06-09 · this version issued 2026-06-09 · v1.0

1. Purpose and scope

The companion note (hdiag-fullcovariance-formulation) reduced the discharge of H-diag to the Bogoliubov-stability inequality $\frac{1}{2}G_*^{-1} \otimes G_*^{-1} \succeq -\mathrm{Hess}\,\Phi_{\mathrm{od}}|_{G_*}$: the condensate-free functional is convex, so the only risk is the condensate-induced off-diagonal Hessian. This note executes the operator's recommended programme – construct $\mathrm{Hess}\,\Phi_{\mathrm{od}}$ from the Bragg-shell coupling and certify the operator-norm domination – and finds that the off-diagonal weight is exactly the ADDITIVE ENERGY of the condensate Bragg set, reducing RES-1 to the DR-2 / R-026 machinery. Scope: second-cumulant / log-det order, the production anchor; BCC verified, the class-wide certificate open.

2. The off-diagonal Hessian from the Bragg-shell coupling (Math428)

For a condensate $\phi_A = A\sum_{j=1}^{N} e^{ik_j x}$ ($|k_j| = q_0$), the Gibbs–Bogoliubov second-order off-diagonal free-energy share is (Math428 §3)

$$\Delta F_{\mathrm{od}}(A) = -\tfrac{1}{4}\sum_{Q\neq 0} W(Q)^2\, B(|Q|), \qquad B(|Q|) = \int \frac{d^3q}{(2\pi)^3}\, G_d(q)\, G_d(q{+}Q),$$

$$W(Q) = (3u + 30vM_R)\, A^2\, p_2(Q) + 5v\, A^4\, p_4(Q) = 3u_{\mathrm{eff}}\, A^2\, p_2(Q) + 5v\, A^4\, p_4(Q),$$

where the leading coupling carries the dressed quartic $u_{\mathrm{eff}} = u + 10vM_R$ (§3 of the companion note: $u_{\mathrm{eff}} = +2.685 > 0$) and the PAIR-STRUCTURE FACTOR

$$p_2(Q) = \#\{(i,j) :\, k_i + k_j = Q\}.$$

$B(|Q|) > 0$ is the convergent bubble (entropy/log-det curvature; $B(0) = -M'(\hat{r})$), so $\mathrm{Hess}\,\Phi_{\mathrm{od}}$ is the negative-definite quadratic form $-\frac{1}{4}\sum_Q(\cdot)^2 B$ and the stability inequality is the ratio

$$\boxed{R(A) = \frac{|\Delta F_{\mathrm{od}}(A)|}{\Delta F_{\mathrm{diag}}(A)} = \left\|E^{-1/2}\,\mathrm{Hess}\,\Phi_{\mathrm{od}}\,E^{-1/2}\right\| < 1.}$$

3. The off-diagonal weight IS the additive energy (exact)

The leading off-diagonal weight is $\propto p_2(Q)$, and its ℓ^2 mass over transfers is, identically,

$$\sum_Q p_2(Q)^2 = \#\{(i,j,k,l) :\, k_i + k_j = k_k + k_l\} = E_+(\{k_j\}),$$

the ADDITIVE ENERGY of the Bragg set. (Verified for the BCC first shell: $N = 12$ $\{110\}$-type vectors, $\sum_Q p_2(Q)^2 = 540 = E_+$ by direct quadruple count; $p_2(0) = N = 12$ reproduces Math428.) Hence

$$|\Delta F_{\mathrm{od}}(A)| \le \tfrac{1}{4}(3u_{\mathrm{eff}}A^2)^2\, B_{\max} \sum_{Q\neq 0} p_2(Q)^2 + (\text{quartic}) \le \tfrac{9}{4}u_{\mathrm{eff}}^2 A^4\, B_{\max}\, E_+, \qquad B_{\max} = \max_{Q\neq 0} B(|Q|).$$

The off-diagonal Hessian mass is bounded by the additive energy of the condensate momentum set – the SAME object as the carrier-richness / additive energy of DR-2.

4. Reduction to the R-025/R-026 machinery (class-wide)

R-025 (Lemma A) gives $E_+ \le (1 + T'(\{k_j\}))N^2$ with T' the maximal sum-circle occupancy; R-026 gives $T' \ll_\varepsilon R^\varepsilon$ for any crystallographic (lattice) shell. (BCC check: $E_+ = 540 \le (1 + T')N^2 = (1 + 4)\cdot 144 = 720$, $T' = 4$, $E_+/N^2 = 3.75 = O(1)$.) The diagonal positive part is $\Delta F_{\mathrm{diag}} \ge c_2 A^2$ with $c_2 = n\, r_R = \frac{N}{2} r_R$ (Math428: $c_2 = 1.827$ for BCC). Imposing the operating-intensity constraint $A^2 \sim I/N$ (fixed condensate intensity I spread over N modes),

$$R(A) \;\lesssim\; \frac{\frac{9}{4}u_{\mathrm{eff}}^2 A^4 B_{\max} E_+}{\frac{N}{2} r_R A^2} \;\sim\; \frac{u_{\mathrm{eff}}^2\, B_{\max}}{r_R}\,\frac{A^2 E_+}{N} \;\sim\; \frac{u_{\mathrm{eff}}^2\, B_{\max}}{r_R}\,\frac{I\, E_+}{N^2} \;\lesssim\; \frac{u_{\mathrm{eff}}^2\, B_{\max}}{r_R}\, I\,(1 + T'),$$

using $E_+/N^2 \le 1 + T'$. For the lattice class $T' \ll_\varepsilon R^\varepsilon$, so R is bounded by the operating intensity I times a SUBPOLYNOMIAL-in-N factor: the off-diagonal stability is controlled CLASS-WIDE by the additive-energy bound, not pattern-by-pattern. This is the same structural mechanism by which R-026 discharged H-ADM-COH.

5. BCC verification and status

Recasting Math428's continuum-anchored verdict as the stability ratio (worst case $\rho = 1$, no resummation credit):

$$R(0.02){=}0.118,\; R(0.05){=}0.602,\; R(0.08){=}0.916,\; R(0.11){=}0.770,\; R(0.14){=}0.510,$$

so $R < 1$ across $[0.02, 0.14]$ (worst 0.916 at $A = 0.08$), and the small/large-A analytic bounds (Math428 §3bis: $\Delta F \sim c_2 A^2 > 0$ as $A \to 0$; sextic-dominated for $A > 0.14$) cover all $A \ge 0$. The small-A scaling $R = O(A^2)$ (off-diagonal $O(A^4)$ vs diagonal $O(A^2)$) is confirmed: $R(0.02) = 0.118 \ll R(0.14) = 0.510$.

T4 STRONG EVIDENCE. Proved (exact): the off-diagonal weight identity $\sum_Q p_2^2 = E_+$. Verified: BCC $R < 1$. Reduced: class-wide $R \lesssim u_{\mathrm{eff}}^2 B_{\max} r_R^{-1} I(1 + T')$, controlled by R-025/R-026. **OPEN (the residual):** the constant-pinned class-wide certificate ($B_{\max}$, ρ, the diagonal lower constant, and the full – not leading – Hessian) for an unconditional H-diag discharge; and its beyond-second-cumulant extension (RES-5). No tier flip: B2-PROPA-HLAYER T6, B1-RH-ENUM T6 on {H-LAYER}.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

(α) *"$R = 0.916$ is thin – the BCC off-diagonal nearly destabilises."* VALID, documented: 0.916 is the WORST case $\rho = 1$ (no resummation); the calibrated $\rho = 0.415$ gives a far safer ratio, and the thin point is the mid-window $A \approx 0.08$, with the $A \to 0$ ($R = O(A^2)$) and large-A (sextic) ends comfortably stable. Thin but < 1, and the operating amplitude $A^2 \sim I/N$ is small.

(β) *"The identity $\sum p_2^2 = E_+$ is exact, but does it CONTROL the Hessian with the right constants?"* VALID-with-mitigation: the identity and the Lemma-A bound are exact (T7 inputs); the REDUCTION uses $B(Q) \le B_{\max}$ and a diagonal lower bound, so the SCALING (subpolynomial in N, linear in I) is robust while the CONSTANT is not yet pinned – hence T4, not a discharge. The leading-p_2 truncation drops the $5vA^4p_4$ term (higher order in A).

(γ) *"This does not discharge H-diag."* DISMISSED (by design): T4 STRONG EVIDENCE; BCC verified + the additive-energy reduction; the class-wide constant certificate is the named residual.

Quantitative sanity check (mandatory): *identity* – $\sum_Q p_2^2 = 540 = E_+$ exactly (direct quadruple count); *magnitude* – $E_+/N^2 = 3.75 = O(1)$, consistent with R-025 ($\le 1 + T' = 5$); *sign* – $\Delta F_{\text{od}} < 0$ (off-diagonal LOWERS F, the destabilising direction), dominated by $\Delta F_{\text{diag}} > 0$; *limit* – $R = O(A^2) \to 0$ as $A \to 0$ ($R(0.02) = 0.118$); *consistency* – BCC worst $R = 0.916 < 1$ reproduces the Math428 PASS.

7. Result footer

```
Result ID:                B2-PROPA-HLAYER / H-diag discharge II (off-diagonal
                          Hessian = additive energy; RES-1 reduced to R-026)
Precise statement:        (1) EXACT: the off-diagonal Bogoliubov weight obeys
                          sum_Q p_2(Q)^2 = E_+({k_j}) (additive energy of the
                          Bragg set; verified 540=540 for BCC N=12). (2) Hence
                          |Delta F_od| <= (9/4) u_eff^2 A^4 B_max E_+, so the
                          class-wide off-diagonal Hessian is bounded by R-025
                          (E_+<=(1+T')N^2) / R-026 (lattice T'<<R^eps); with
                          A^2~I/N, R <~ u_eff^2 B_max r_R^-1 I (1+T') (subpolynomial
                          in N). (3) VERIFIED: BCC stability ratio R=|off-diag|/diag
                          < 1 (worst 0.916 at A=0.08), = ||E^-1/2 Hess Phi_od
                          E^-1/2|| < 1, recasting Math428.
Scope:                    Second-cumulant / log-det order, anchor mu^2; leading-p_2
                          coupling; BCC verified, class-wide reduction T4.
Dependencies:             Math428 (Bloch off-diagonal share, W(Q), B(Q), continuum
                          table); R-025/R-026 (additive energy E_+<=(1+T')N^2,
                          lattice T'); companion hdiag-fullcovariance-formulation
                          (F0 convex, u_eff=+2.685).
Evidence grade:           ANALYTIC (identity + bound) + EXECUTED
                          (hdiag_offdiag_additive_energy.py 6/6) + INHERITED
                          (Math428 continuum table).
Reproduction command:     python codes/vacuum/hdiag_offdiag_additive_energy.py
Expected output:          sum p_2^2 = E_+ = 540; E_+<=720; BCC R worst 0.916<1;
                          R(0.02)=0.118; 6/6 PASS.
Falsification gate:       A Bragg set with sum_Q p_2^2 != E_+ (would void the
                          identity); a pattern with R>=1 at the operating intensity
                          (off-diagonal NOT dominated); or E_+ exceeding (1+T')N^2.
Tier before / after:      No tier flip. B2 T6, B1 T6 on {H-LAYER} unchanged.
                          H-diag off-diagonal sector: reduced to an additive-energy
                          bound, BCC verified (T4 strong evidence).
No-overclaim statement:   This does NOT discharge H-diag or H-LAYER and does NOT
                          promote B1. The identity sum p_2^2=E_+ and the Lemma-A
                          bound are exact; the class-wide off-diagonal DISCHARGE
                          needs the constants (B_max, rho, diagonal lower bound,
                          full Hessian) pinned -- the named residual -- plus the
                          RES-5 beyond-second-cumulant extension. BCC R=0.916 is
                          the thin worst case.
```

Next required action: Pin B_max and the diagonal lower constant to convert the subpolynomial reduction into a class-wide certificate; include the 5vA^4 p_4 sub-leading term; then attempt the lattice-class H-diag discharge (mirroring the H-ADM-COH discharge via R-026).